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Mattia Rengo

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ABOUT ME

Registered Information Engineer in Italy, Section A, having passed the State Examination. DevSecOps Engineer in R&D, fully remote since April 2023. Contributions to AI platforms for legaltech, the public sector and healthcare research: Kubernetes, secure software delivery, Python backend services and MLOps. Applied research, publications, open source and knowledge sharing. English C1.

WORK EXPERIENCE

DEVSECOPS ENGINEER – EUSTEMA S.P.A. – 07/2025 – Current – ITALY

R&D of AI platforms for legaltech, the public sector and healthcare. Responsible for the CI/CD infrastructure and deployment design deliverables for an AI agent research platform.
- Designed shared templates and GitLab CI pipelines with Argo CD GitOps promotion, progressively automating software delivery while retaining manual approval for production.
- Integrated SBOMs, vulnerability scanning and Cosign signature and attestation verification before promoting the same image across environments.
- Operated AI infrastructure on Kubernetes: LLM serving with KubeAI and vLLM, GPU allocation and persistent storage, MLflow migrations and observability services.
- Developed FastAPI microservices and shared libraries for document management, configuration, audit and authorization. Standardized logging, tracing, probes and graceful service shutdown.
- Developed and deployed a Kubernetes operator using CRDs and Kopf to manage AI agents in development environments, with functional testing and a deployment demonstration.
- Authored architecture studies and deliverables covering CI/CD, deployment, resilience and sizing. Trained teams through guides and workshops, enabling independent pipeline execution.
- Contributed to healthcare AI research by planning reproducible infrastructure and DevSecOps/MLOps workflows.

DEVOPS DEVELOPER – EUSTEMA S.P.A. – 04/2024 – 07/2025 – ITALY

Infrastructure and automation for the R&D platform, AI document services and support for cloud releases in public-sector projects.
- Built Coder remote development environments on Kubernetes with Terraform templates, Docker and GPU workspaces, and developer onboarding.
- Developed document services and delivery pipelines for a legaltech AI platform. Improved ingestion through load testing and parallelization, and automated GitLab repository backups and exports.
- Designed and developed a framework to evaluate deployed RAG applications, contributing to applied research and its technical report as first author.
- Managed Kubernetes services, storage, monitoring and releases on premises and in the cloud, including GKE and AWS EKS test environments for public-sector projects.

SOFTWARE DEVELOPER – EUSTEMA S.P.A. – 04/2023 – 04/2024 – ITALY

Moving an AI document platform from research to production on Google Cloud, alongside contributions to rehabilitation medicine research.
- Containerized services and prepared Kubernetes manifests for an AI platform serving a Lombardy regional public-sector organization, contributing to its production deployment on Google Kubernetes Engine (GKE).
- Ran Bandit SAST analysis and load tests. Resolved ingress and probe issues and maintained release images and platform backups.
- Researched and documented HL7 FHIR interoperability and data exchange for rehabilitation medicine research projects.

COMPANY INTERNSHIP AND EXPERIMENTAL CLOUD BURSTING THESIS – EUSTEMA S.P.A. – 10/2022 – 04/2023 – ITALY

Experimental company thesis on automatically expanding Kubernetes across on-premises infrastructure and private clouds in a hybrid multi-cloud environment.
- Built two OpenStack cloud environments from scratch and connected them to an on-premises cluster to experiment with Kubernetes K3s spanning a hybrid multi-cloud architecture.
- Developed a cloud bursting proof of concept and evaluated automatic resource expansion using containerized services and load tests.

EDUCATION AND TRAINING

09/2020 – 28/04/2023 Perugia, Italy

MSC IN COMPUTER AND ROBOTICS ENGINEERING University of Perugia

Data Science curriculum. Grade: 110/110 cum laude. Experimental thesis: building two OpenStack clouds and integrating them with an on-premises cluster for Kubernetes cloud bursting in a hybrid multi-cloud environment.

Website https://ing.unipg.it/ | **Final grade** 110/110 cum laude | **Thesis** Approcci per il cloud bursting basati su Kubernetes

09/2014 – 2020 Perugia, Italy

BSC IN COMPUTER AND ELECTRONIC ENGINEERING University of Perugia

Grade: 99/110. Thesis on designing and analyzing a data center network, with infrastructure simulation using Huawei eNSP.

Website https://ing.unipg.it/ | **Final grade** 99/110 | **Thesis** Architettura ed analisi di una rete per data center tramite eNSP

PROFESSIONAL REGISTRATION

Order of Engineers of Perugia - Section A

Registered Information Engineer in Italy, Order of Engineers of the Province of Perugia, Section A, no. 4540, since 1 July 2024. State Examination passed at the University of Perugia, second session 2023.

TECHNICAL SKILLS

Areas of practice

Kubernetes and cloud: Kubernetes, GKE, AWS, Docker, Helm, Kustomize, Linux, NFS, MetalLB, NGINX Ingress, CRD, Kopf.
CI/CD and GitOps: GitLab CI, Argo CD, promotion, rollback, multi-architecture builds, Terraform for Coder workspaces.
DevSecOps: Syft, Grype, Cosign, SBOM, signatures, attestations, Bandit SAST, Keycloak, JWT, RBAC.
MLOps and LLMOps: MLflow, KubeAI, vLLM, Ollama, NVIDIA GPU, MinIO, Langfuse, RAG, DeepEval.
Backend and data: Python, FastAPI, Pydantic, Bash, MongoDB, PostgreSQL, Elasticsearch, Neo4j, NATS, Temporal.
Observability and collaboration: OpenTelemetry, JSON logging, probes, load testing, documentation, onboarding.

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
FRENCH	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DRIVING LICENCE

Driving Licence: B

CERTIFICATIONS AND COURSES

Qualifications earned

Microsoft Certified Azure Fundamentals AZ-900 (2023). Huawei HCIA-Routing & Switching (2020). Cambridge First Certificate in English (2013), English C1. AI Agents Fundamentals course - Hugging Face (2025).

PROJECTS

Publications

First author of a RAG evaluation technical report (arXiv, 2025). Coauthor of a Kubernetes cloud bursting paper (IEEE OJCOMS, 2024).
arXiv 2504.07803 | IEEE DOI

Open source contributions

Merged Coder, KubeAI and CheshireCat contributions on Helm, networking and containers. Personal Docker projects, including money4band.
Coder #16032 | KubeAI #362 | CheshireCat #857 | GitHub MRColorR

INNOVATION AND KNOWLEDGE SHARING

Cybersecurity innovation - 2023

Special recognition for the most innovative project in Eustema's internal challenge (2023). Originator and coordinator of an AI-based automated penetration testing proof of concept, exploring GANs and MulVAL and presented to the company jury.

Presentations and professional development

Presented professional roles and enterprise technologies at the University of L'Aquila Career Day (2024) and the University of Perugia Career Day (2025). Attended Red Hat Summit: Connect in Rome (2024 and 2025) and the DLT Workshop in Perugia (2025).

I authorise the processing of my personal data in this CV under Article 13 of Legislative Decree 196/2003 (Personal Data Protection Code) and Article 13 of Regulation (EU) 2016/679 (General Data Protection Regulation).